

Atishaya Maharjan

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EDUCATION & HONOURS

BSc. Joint Honours: Computer Science & Mathematics, Minor in Statistics Aug. 2022 – Present
University of Manitoba *Winnipeg, Canada*

International Undergraduate Student Scholarship | Dean's Honour Roll

RESEARCH EXPERIENCE

Undergraduate Researcher – Algorithms & Computational Geometry Mar. 2024 – Present
Geometric, Approximation, and Distributed Algorithms (GADA) Lab *Winnipeg, Canada*

- Developing faster exact and approximation algorithms for fuel-efficient collaborative delivery using dynamic programming, exchange-point geometry, and structural properties of optimal delivery sequences.
- Developed an approximation algorithm for Hamiltonian-path optimization in spatially embedded graphs , with visualization and randomized testing infrastructure covering **100+** synthetic instances.
- Improved the performance of a noncrossing spanning-tree research library by **30%** and introduced stronger **Python** testing, documentation, and development practices.

Undergraduate Researcher – Combinatorics & Graph Theory Jan. 2024 – Jul. 2025
Department of Mathematics, University of Manitoba *Winnipeg, Canada*

- Established a coefficientwise inequality underlying the Hurlbert–Kamat property for perfect m -ary trees  and derived a closed-form expression for their independence number.
- Applied generating functions, polynomial recurrences, induction, and coefficientwise analysis to develop the mathematical argument.
- Implemented **SageMath** and **Python** experiments across **1,000+** instances to test conjectures, verify symbolic identities, and validate intermediate results.

Machine Learning Researcher May 2025 – Present
Terrabyte Machine Learning Research Group *Winnipeg, Canada*

- Generalized Graph Neural Diffusion architectures to hypergraphs, implementing **4** Hypergraph Neural Diffusion variants using **PyTorch** and **PyTorch Geometric**.
- Built a reproducible **MLflow** pipeline tracking **25+** experiments across **5+** datasets.
- Tracked configurations, metrics, and evaluation results to support reproducible model comparison and analysis.

PROFESSIONAL EXPERIENCE

Software Engineer Mar. 2024 – Present
LG Electronics *Englewood Cliffs, USA (Remote)*

- Built features for a B2B platform serving **1,000+** vendors using **Next.js**, **Redux**, and **TypeScript**, contributing to a **25%** increase in user engagement.
- Built analytics and custom tagging frameworks used by **6+** product teams to support product measurement and decision-making.
- Developed global-navigation search work, including an average search-latency reduction of **35 ms**.

Software Engineer | Vibecore – Cloud Infrastructure & CI/CD

- Automated **GitHub Actions** CI/CD workflows to **AWS EC2** instances for application deployment.
- Managed **S3** buckets, **DynamoDB** integrations, and **IAM** permissions for cloud-connected workflows.
- Provisioned and maintained **EBS/EFS** storage for EC2-hosted services and deployment infrastructure.

Software Engineer Jun. 2025 – Jan. 2026
Hoek Insights *Amsterdam, Netherlands (Remote)*

- Built a **Python** package embedding **Three.js** visualizations in **Jupyter Notebooks** via **React** renderers, cutting render time by **38%** and bundle size by **25%**.
- Designed a **FastAPI/PostgreSQL** MVP with **18+** endpoints and **10** workflows for a marketing platform.
- Built integration-test coverage to **95%** and maintained below **200 ms** median API latency in staging.

Software Engineer Aug. 2019 – Apr. 2024
Timeero *Phoenix, USA (Remote)*

- Led frontend development of a mileage-tracking application using **TypeScript**, **React**, **Vite**, and **Mapbox**.
- Supported **1,000+** active users across production mileage-tracking and time-capture workflows.
- Implemented **15+** geospatial mapping features for segment mapping and time-capture accuracy.

SELECTED RESEARCH AND TECHNICAL PROJECTS

Brainstorm | *Authorization-Aware RAG, Data Systems*

- Built a role-aware knowledge-management platform integrating **Jira**, **PostgreSQL**, **Redis**, and **Pinecone** through retrieval-augmented generation.
- Implemented source grounding and role-based access control across connected organizational data systems.
- Built multi-model routing with fallback policies balancing latency, reliability, and response quality.

WinLocate | *Applied Data Engineering, Mapping*

- Integrated **Socrata** open data into an applied data platform for city and relocation analysis.
- Built city comparisons, interactive zoning maps, budget tools, and relocation decision support workflows.
- Won **1st place** among **15** teams at the Winnipeg Open Data Datathon 2026.

Get Good Grades | *Electron, React, TypeScript, YAML*

- Built a decentralized flashcard viewer that ingests `.yaml` flashcard files from anywhere in the file system.
- Gave users full control over flashcard content and scheduling logic instead of relying on centralized storage.
- Presented the project at DevHacks 2026.

RESEARCH AWARDS

S.U.R.E. Poster Competition – Third Place, Computational Science Category

Aug. 2024

- Received third place for the research poster “*Minimizing Maximum Interference in 1-Dimensional Networks*” at the University of Manitoba.

Winnipeg Open Data Datathon – 1st Place among 15 teams

2026

CSSA Capture the Flag – 1st of 41 teams (2026), 2nd of 25+ teams (2025)

.devHacks – Most Chaotic Evil Hack among 20+ teams

TECHNICAL SKILLS

Programming: Python, C++, C, TypeScript, JavaScript, Java, R, SQL, Bash, SageMath

Scientific Computing & ML: PyTorch, PyTorch Geometric, NumPy, SciPy, scikit-learn, pandas, MLflow

Systems & Cloud: FastAPI, PostgreSQL, Redis, MongoDB, DynamoDB, Pinecone, AWS EC2, S3, IAM, Docker, GitHub Actions

Visualization: Jupyter, Three.js, Mapbox, React

Research Tools: Git, Linux, LaTeX, Beamer, RAG, Model Context Protocol

TEACHING AND LEADERSHIP

CS Help Center Leader & Teaching Assistant | *University of Manitoba*

Jan. 2025 – Apr. 2025

- Mentored **100+** students across first- through third-year computer science courses and graded **150+** assignments for Analysis of Algorithms (COMP 3170).

Coding Kickoff Facilitator | *University of Manitoba*

Feb. 2025 – Mar. 2025

- Facilitated workshops that taught **IDE setup** and **Git** to **50+** first-year university students.

Events Executive | *.devClub (University of Manitoba)*

Aug. 2023 – Jul. 2024

- Co-led workshops, hackathons, academic review sessions, and community events for the Computer Science Department at the Faculty of Science.

Audio Visual Team Member | *Bethesda Church*

Jul. 2023 – Present

- Operated A/V equipment for weekly sermons and live events, ensuring high-quality audiovisual experiences for attendees.